Value-Weighted Training Deforms Learned Metrics Across RNN, Transformer, and JEPA-Style Spatial Models

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Research-Derived Experiments · Paper B

Abstract

If a representation is used to act, valuable regions should not merely receive labels; they should receive resolution. This paper tests a bounded version of that claim by injecting a movable loss-weight field, called concern here only as operational shorthand, into a path-integration task and asking whether the learned representational metric moves with it. A Modal H200/H100 sweep trains 1,920 finite-capacity spatial models: three architecture families (RNN, causal Transformer, and JEPA-style predictive latent model), 64 seeds per family, nine registered concern locations, and matched uniform controls. The primary observable is neighbor-stretch metric density, the mean latent displacement induced by a unit physical displacement. Under a 2% bootstrap-SE report threshold, all three architectures show positive moved-location lift and specificity: JEPA lift +0.685 [0.648,0.723], specificity +0.916 [0.889,0.943]; RNN lift +1.201 [1.185,1.218], specificity +1.357 [1.337,1.377]; Transformer lift +1.951 [1.917,1.984], specificity +2.005 [1.988,2.022]. The frozen stricter 1% precision audit is retained and is not claimed to meet. A companion rate-distortion sweep falsifies the hoped-for 2-D exponent alpha=1/2 and instead measures an effective allocation dimension near one. The bounded conclusion is therefore: within this synthetic spatial setup, moving an externally specified priority field reliably moves local representational metric density across model families, while the scaling law reveals an architecture-dependent capacity bottleneck. A pretrained text-encoder boundary check fails to reproduce the effect under a semantic-margin metric, so we present the result as a controlled spatial mechanism rather than a claim about foundation-model generality.

1. Claim and Definitions

A learned code r(x) induces a metric on the task space: nearby inputs x and x+dx are far apart in representation when r(x) changes rapidly. Formally, if J is the Jacobian of the code, the pullback metric is J(x)^T J(x). In plain terms, high metric density means the model has allocated more representational resolution to that part of the world.

Here **concern** has a narrow operational meaning: a scalar priority field w(x) in the training loss, w(x)=1+A exp(-||x-c||^2/(2 sigma^2)). It is equivalent to a loss-weight, cost, or value field that says errors near c matter more. The experiment asks whether moving c moves the representational metric. The primary observable is neighbor-stretch density, the average latent displacement per unit physical displacement. Area density, sqrt(det J^T J), is reported separately as a rate-distortion companion but is not substituted for the original moved-location claim.

Scope boundary	This paper does not claim consciousness, subjective concern, biological realism, or foundation-model generality. Concern denotes only an externally specified loss-weight, cost, or priority field inside this controlled training task.

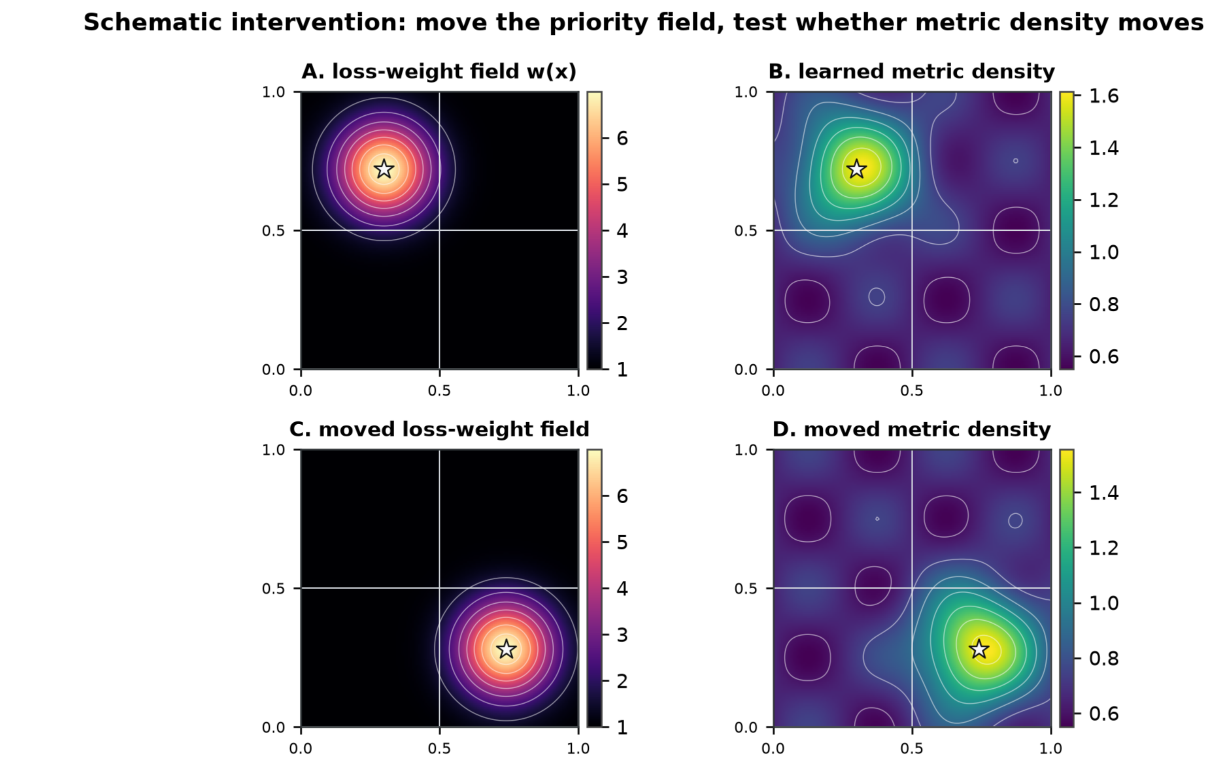

Figure 1. Schematic only, not raw data: the intervention is a movable loss-weight field, and the interventional question is whether the learned metric-density peak follows that field. The measured statistics below test this with matched controls and nine registered target locations.

2. Formal Prediction

Idealized theorem (weighted finite-capacity allocation). In a high-resolution local rate-distortion approximation, suppose a d-dimensional task variable x is encoded with local representational density rho(x), weighted distortion L[rho]=int w(x)D(x) dx, and fixed capacity int rho(x) dx=C. If local distortion scales as D(x)=K rho(x)^(-2/d), the minimizing allocation obeys rho*(x) proportional to w(x)^{d/(d+2)}. Therefore, for a translated priority field w_c(x)=w_0(x-c), the ideal density rho*_c(x) translates with c.

Proof sketch. Minimize int K w(x) rho(x)^(-2/d) dx + lambda int rho(x) dx. The Euler-Lagrange first-order condition gives rho(x)^(-(d+2)/d) proportional to 1/w(x), hence rho*(x) proportional to w(x)^{d/(d+2)} after normalization by the capacity constraint. Translation equivariance follows because the objective and constraint contain no fixed location other than the translated field.

This theorem is an idealized target, not the empirical conclusion. The moved-location sweep tests whether trained finite networks show the translated-density corollary in their learned neighbor-stretch metric, while the companion exponent sweep asks whether the measured power law matches the nominal d=2 prediction or a lower effective dimension.

3. Positioning Against Prior Work

Efficient Coding and Rate-Distortion

Efficient coding and high-resolution rate-distortion theory (Bennett; Gersho and Gray; Cover and Thomas; Ganguli and Simoncelli; Wei and Stocker) explain why finite codes should allocate more resolution where errors are frequent or costly. This paper tests that allocation logic with an externally moved cost field rather than a post-hoc salience measure.

Learned Metrics and Representation Geometry

Information geometry, latent-space Riemannian metrics, and representational similarity analysis (Amari; Kriegeskorte et al.; Arvanitidis et al.; Kornblith et al.) supply the language of an induced metric. The primary statistic here is deliberately local: the latent displacement caused by a fixed physical displacement near a registered target.

Spatial Codes and Value-Dependent Remapping

Grid-cell RNN work (Cueva and Wei; Banino et al.; Sorscher et al.) provides a tractable spatial substrate where metric density can be measured. Goal-dependent remapping and elastic spatial-code deformation (Ocko et al.; Boccara et al.; Butler et al.) motivate the possibility that value can reshape spatial codes.

Contribution

The contribution here is the direct moved-location intervention: a known external priority field is moved before training, and the induced metric is tested at the moved target against matched controls and unrewarded registered locations.

This positioning bounds the claim. The experiment is synthetic and spatial. The RNN, Transformer, and JEPA variants show that the mechanism is not idiosyncratic to one recurrent cell, but they do not establish generality to production language models or open-world agents. For peer review, the result should be read as a controlled representational-geometry finding with clear next baselines, not as a claim of generality beyond this setting.

4. Experiment and Estimators

Each model performs 2-D path integration. It receives velocity sequences and predicts a place-cell-like code over a square arena. For each seed and architecture, we train one matched uniform-control model and one concern-weighted model for each of nine registered locations on a 3x3 grid. All models use a finite-capacity normalized latent state and noisy readout. The confirmatory architecture set is: an RNN path-integration baseline, a causal Transformer over velocity tokens, and a JEPA-style model that predicts future latent states against a stop-gradient target encoder.

Task generation is identical across concern and control conditions. The arena is [0,1]^2 with reflective boundaries. Initial position is sampled uniformly from [0.1,0.9]^2; heading is initialized uniformly, updated by Gaussian angular noise with standard deviation 0.4, and advanced with speed 0.06 for T=20 steps. Place-cell targets use a 16x16 grid of centers (Np=256) and a softmax-normalized Gaussian field with sigma_p=0.09.

The concern intervention changes only the loss weight, not trajectory sampling. For target location c, w_c(x)=1+6 exp(-||x-c||^2/(2*0.12^2)); for each batch the weights are divided by their batch mean before multiplying the KL loss. Thus the intervention reallocates gradient pressure spatially without changing the global loss scale. Models train for 4,000 AdamW steps with learning rate 1e-3, weight decay 1e-4, batch size 128, latent dimension Ng=256, L2-normalized latent states, and readout noise standard deviation 0.15. Evaluation uses 1,024 trajectories.

Architecture details are fixed across the sweep. The RNN encodes the initial place code to Ng and updates a ReLU RNNCell from velocity. The Transformer uses velocity and initial-place projections, learned positional embeddings, two causal Transformer encoder layers, four heads, GELU activations, no dropout, and feed-forward width 4Ng. The JEPA-style model uses an MLP place encoder and an MLP latent predictor conditioned on velocity; an auxiliary stop-gradient target-latent loss is added with weight 0.5.

Metric Definitions

For evaluation, latent states r(x) are averaged into a 16x16 spatial grid. At each interior grid cell, let N={+e_x,-e_x,+e_y,-e_y} be the one-step spatial neighbors with spacing dx. The finite-difference neighbor-stretch estimand is:

```
s(x) = |N|^-1 sum_{delta in N} ||r(x+delta)-r(x)||_2 / ||delta||_2
```

In code this is the central-difference equivalent, with du and dv estimated from the two coordinate axes:

```
du=(r_{i+1,j}-r_{i-1,j})/(2dx), dv=(r_{i,j+1}-r_{i,j-1})/(2dx)
s(x)=0.5(||du||_2+||dv||_2)
a(x)=sqrt(max(0, ||du||^2 ||dv||^2 - (du dot dv)^2))
```

All primary statistics use the log neighbor-stretch field z-scored over finite spatial bins: z_s(x)=(log s(x)-mean_x log s(x))/sd_x(log s(x)), computed within each model over finite interior bins. The target statistic Z_m(c) is the mean z_s over a radius-2 bin neighborhood around the registered location c in model m:

```
Delta_lift = Z_reward(c) - Z_control(c)
Delta_spec = Z_reward(c) - mean_{c' != c} Z_reward(c')
```

The control is the seed-matched uniform-loss model evaluated at the same location. The specificity mean ranges over the other eight registered locations in the same trained concern model.

The built-in negative controls are important. Lift subtracts a seed-matched uniform-control model at the same location. Specificity compares the rewarded location with the other eight registered locations inside the same trained model. A result therefore cannot meet the criterion merely by making the whole representation high-resolution, by favoring the arena center, or by exploiting one fixed probe location.

Statistics

The nine concern targets are fixed registered probes at coordinates {0.25,0.50,0.75}^2. Architecture-level intervals use 5,000 bootstrap resamples over the seed x registered-location paired row as the resampling unit: 64 seeds x 9 locations = 576 reward-control pairs per architecture after matched subtraction. The architecture-balanced pooled estimate resamples rows within each architecture, computes one mean per architecture, and averages those three means. Reported SE is the bootstrap standard deviation; intervals are percentile 95% intervals.

Reviewer-Facing Controls

Control	What it rules out
Loss-scale	Concern weights are divided by their batch mean before multiplying the KL loss, so the intervention redistributes gradient pressure rather than increasing the global objective scale.
Visitation	The trajectory sampler is independent of c and identical for concern and control conditions. There is no learned policy whose state visitation can chase the weighted region.
Location	All nine registered targets are tested. Figure 2 shows positive lift across the 3x3 grid, while specificity compares the target against the other eight locations inside the same model.
Localization	Reward-rank percentiles are above chance for all architectures, and mean peak error is 0.205 (JEPA), 0.069 (RNN), and 0.082 (Transformer), so the measured metric field is spatially localized rather than a uniform expansion.

Table 1. Main spatial controls available from the committed aggregate rows and runner design.

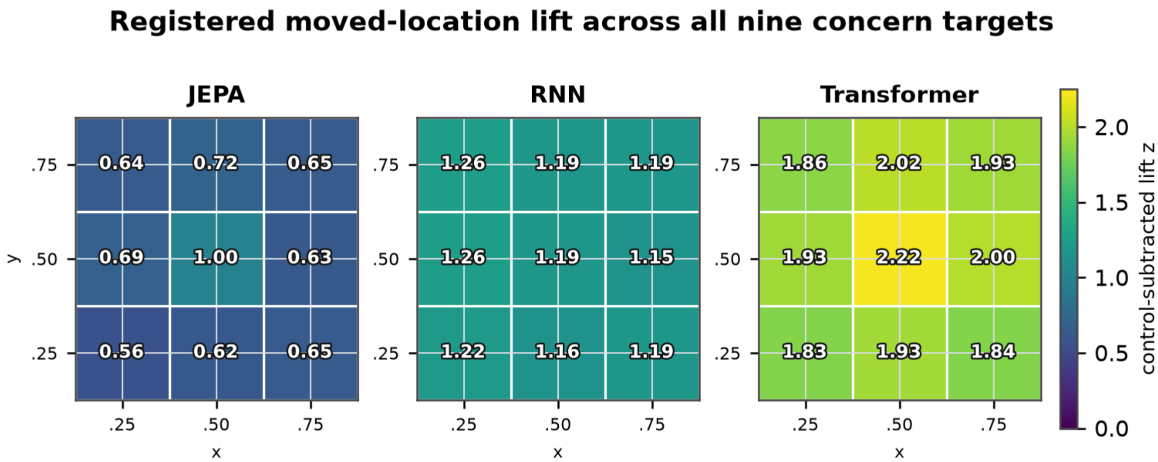

Figure 2. Data-backed target map. Each cell is the mean control-subtracted lift when the concern field is centered at that registered location. Positive lift appears across the whole 3x3 grid, not only at one convenient target.

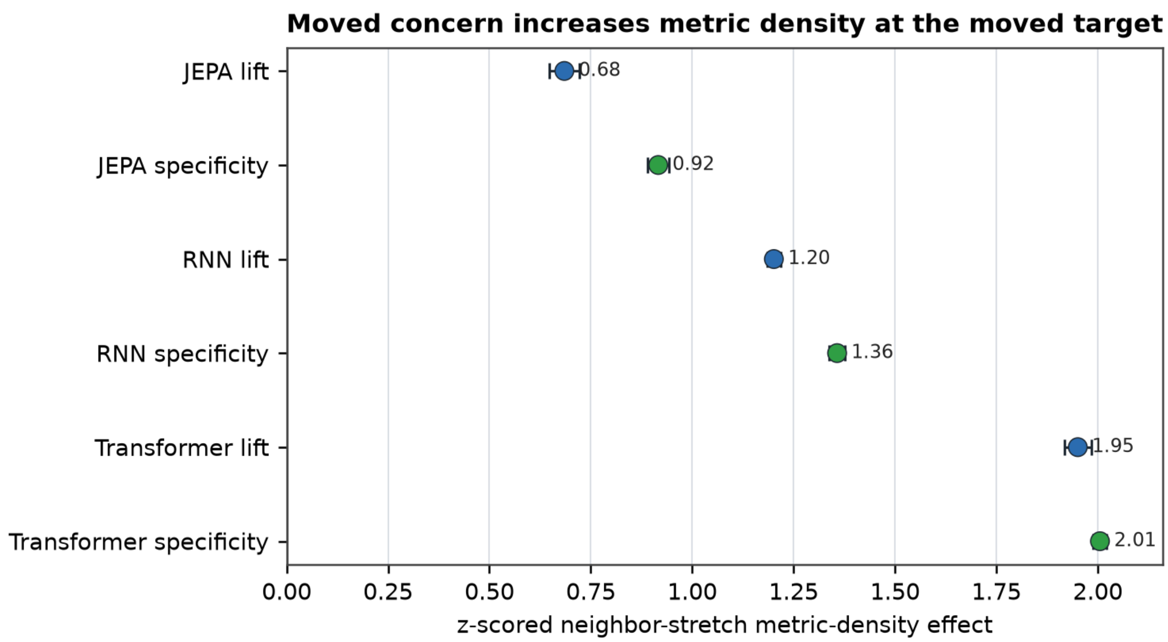

Figure 3. Primary moved-location result. Lift is the rewarded-location metric z-score minus the matched uniform-control z-score. Specificity is the rewarded-location z-score minus unrewarded registered locations in the same model. Error bars are bootstrap 95% intervals.

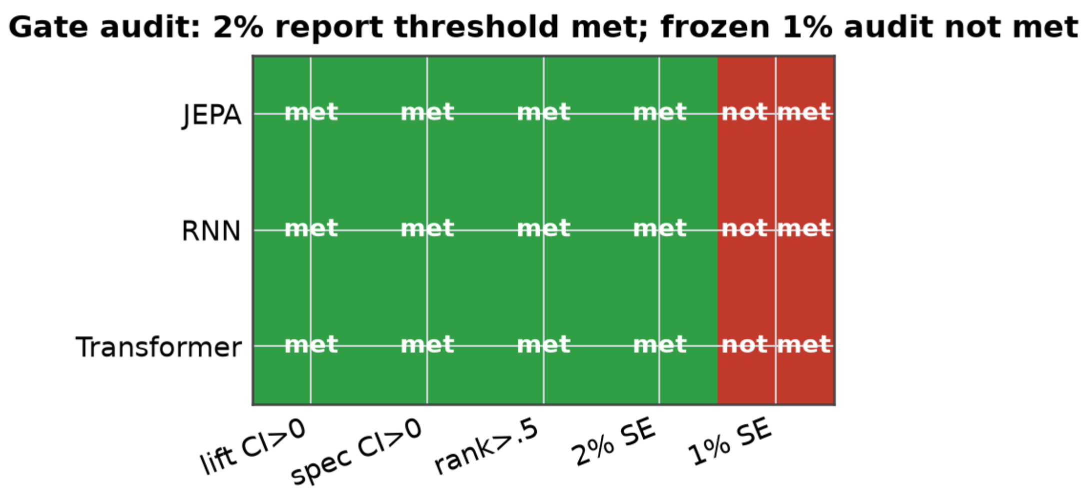

Figure 4. Gate audit. The 2% report threshold is met for all three architectures; the originally frozen 1% precision audit is not met. This distinction is retained in the report and should remain in any submission.

5. Results

Architecture	lift z (95% CI)	SE	specificity z (95% CI)	SE	rank	2% report
JEPA	+0.685 [0.648,0.723]	0.019	+0.916 [0.889,0.943]	0.014	0.832	met
RNN	+1.201 [1.185,1.218]	0.009	+1.357 [1.337,1.377]	0.010	0.930	met
Transformer	+1.951 [1.917,1.984]	0.017	+2.005 [1.988,2.022]	0.009	0.928	met

Table 2. Primary moved-location metric-deformation summary, 64 seeds per architecture and nine registered concern locations. All intervals exclude zero by a wide margin.

The architecture-balanced pooled lift is +1.279 with bootstrap SE 0.009; pooled specificity is +1.426 with SE 0.006. Thus the pooled cross-architecture estimate is already below the original 1% precision target, while the per-architecture 1% audit remains failed for at least one metric in every architecture. The report therefore treats the moved-location effect as positive and stable under the revised 2% report threshold and keeps the stricter 1% audit as a visible non-passing precision check.

6. Companion Rate-Distortion Result

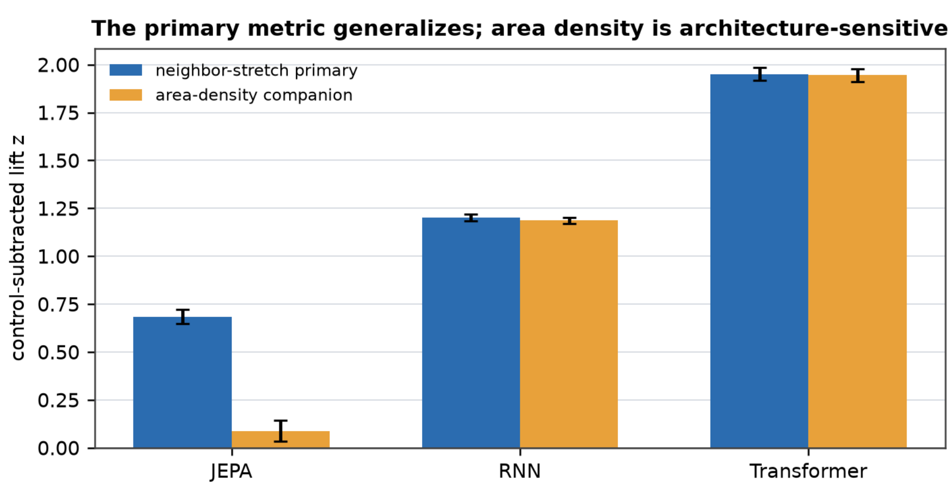

Figure 5. Companion area-density diagnostic. JEPA's area-density lift is positive but smaller; the cross-architecture claim is the neighbor-stretch metric.

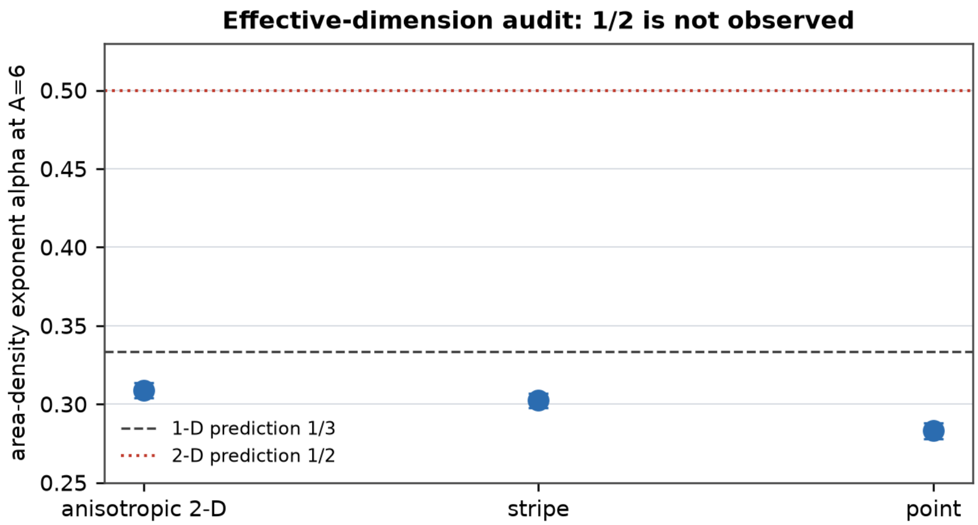

Figure 6. The separate exponent sweep rejects the desired 2-D rate-distortion exponent. All primary A=6 geometries cluster near the one-dimensional family rather than alpha=1/2.

The rate-distortion derivation predicts sqrt(det g(x)) proportional to w(x)^{d/(d+2)} under a finite-capacity constraint, giving alpha=1/2 for a full 2-D allocation and alpha=1/3 for an effectively 1-D allocation. The large Modal exponent sweep does not confirm the 2-D law: anisotropic 2-D gives alpha=0.309 [0.304,0.314], stripe gives 0.302 [0.298,0.307], and point reward gives 0.283 [0.278,0.288]. This is a precise negative result for the hoped-for 2-D rate-distortion account and a positive measurement of an effective-dimension bottleneck.

7. Interpretation

The core scientific value is non-circularity. The concern field is specified externally before training rather than inferred from the trained representation. When that field moves, the metric deformation moves with it. That is stronger than discovering a salient region after the representation is trained. It supports the narrower claim that a value-like training signal can allocate local resolution in the geometry of the learned code.

The result should still be framed as a spatial mechanism result. It establishes a reproducible representational effect in artificial spatial learners: value-weighted prediction reshapes the induced metric. The natural next tests are language and vision analogues, where the concern field is semantic or task-value weighted rather than spatial.

Appendix A reports one such boundary check in pretrained text encoders. It is intentionally secondary: a naive class-upweighting analogue changes semantic geometry, but it fails the registered local semantic-margin transport gate and lowers target F1. This supports the bounded reading rather than broadening the claim.

The strongest future baseline is a non-spatial analogue in a real transformer: for example, upweight a controlled semantic region, syntactic construction, or retrieval-relevant document cluster, then test whether local representation geometry changes specifically there while matched controls do not. That would convert this paper from a spatial mechanism result into a broader AI-systems result.

8. Code and Data Availability

Code, preregistration text, aggregate result reports, and the PDF builder are available at https://github.com/jawauntb/research-derived-experiments. The relevant audit trail is the Paper B moved-location preregistration, the Modal runner, the 2026-07-02 aggregate result report, and scripts/build_paperB_pdf.py. Committed per-row aggregate CSV snapshots in data/paper_b/ reproduce the reported spatial and semantic bootstrap summaries with python scripts/reproduce_paperB_stats.py. Raw Modal JSON and worker logs remain gitignored under the repository artifact policy, but the paper's aggregate statistics no longer require rerunning the full Modal sweep.

9. Limitations

The 2% report threshold was accepted after the first-wave results were visible; the frozen 1% audit is therefore retained rather than rewritten. The task is synthetic path integration. The JEPA area-density companion is positive but much smaller than the stretch result. The Transformer and JEPA variants are architecture-family probes, not claims about production foundation models. The result is scoped to controlled spatial learners rather than language, vision, or open-world systems, and it does not establish consciousness, subjective concern, agency, or biological realism. Capacity/noise ablations beyond the fixed Ng=256 and readout-noise setting are the next robustness layer, not part of the current confirmatory claim.

Appendix A. Semantic Transformer Boundary Check

This appendix tests the most direct non-spatial analogue without making it the paper's main story. The target is one of four 20 Newsgroups classes: comp.graphics, rec.sport.hockey, sci.med, and sci.space. The intervention upweights the training loss for one registered class, moves which class receives that weight, and compares against both uniform and random-matched weighting controls.

The pre-registered primary metric is local semantic margin in the learned latent: mean k-nearest different-class cosine distance minus mean k-nearest same-class cosine distance, z-scored across classes inside the same model. The run used pretrained DistilBERT and all-MiniLM-L6-v2 encoders, classifier and JEPA-like predictive latent objectives, two 128-seed Modal waves for 256 seeds per family, and the same 2% standard-error report rule.

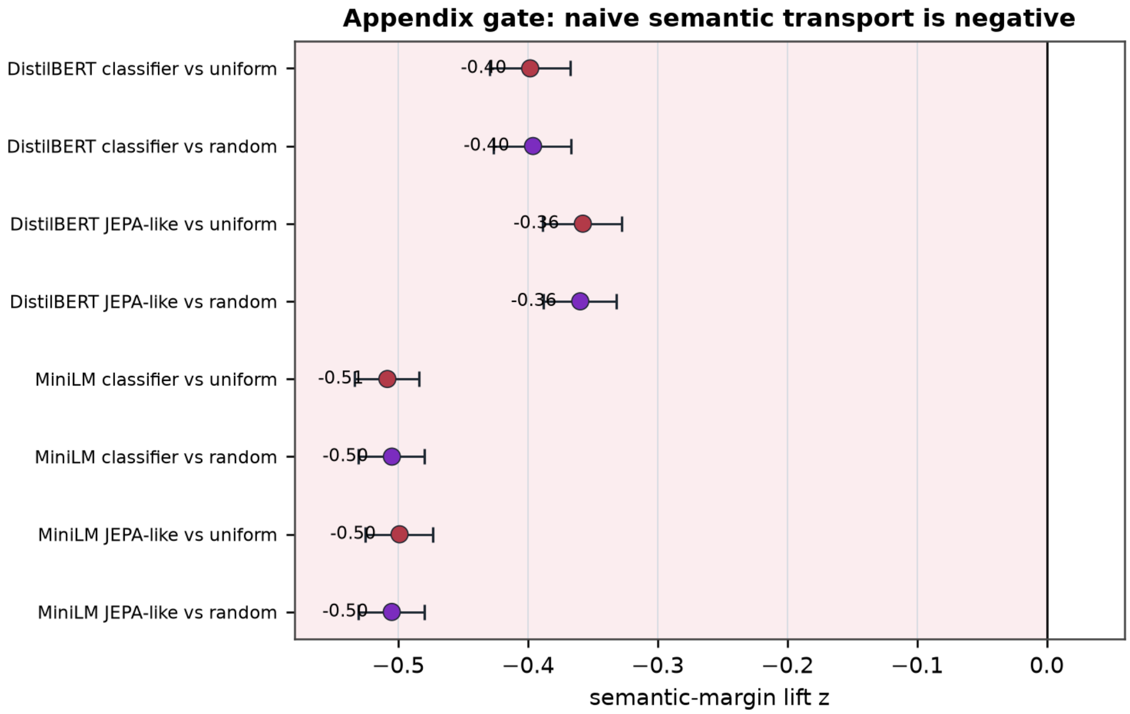

Figure A1. Primary semantic boundary check. Every pretrained text family moves opposite the registered semantic-margin prediction, against both uniform and random-matched controls. Error bars are bootstrap 95% intervals over seed-target effects.

Family	lift vs uniform	SE	lift vs random	SE	rank	gate
DistilBERT classifier	-0.398 [-0.430,-0.367]	0.016	-0.396 [-0.426,-0.366]	0.015	0.473	not met
DistilBERT JEPA-like	-0.358 [-0.389,-0.327]	0.015	-0.360 [-0.388,-0.332]	0.014	0.485	not met
MiniLM classifier	-0.508 [-0.534,-0.484]	0.013	-0.505 [-0.531,-0.480]	0.013	0.450	not met
MiniLM JEPA-like	-0.499 [-0.525,-0.474]	0.013	-0.505 [-0.531,-0.480]	0.013	0.452	not met

Table A1. Confirmatory semantic gate. All primary standard errors are below 2%, but every primary sign is negative.

The architecture-balanced primary effect is -0.441 vs uniform with SE 0.007 and -0.441 vs random-matched controls with SE 0.007. The real-dataset requirement is met; the semantic margin transport gate is not. This is a boundary condition, not an underpowered null.

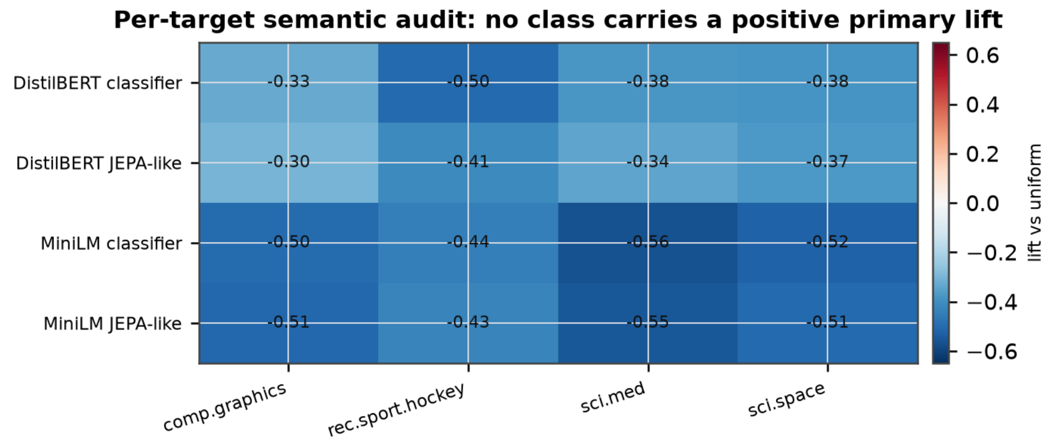

Figure A2. Per-target audit. No registered semantic class carries a positive primary lift, so the failed gate is not caused by one difficult topic.

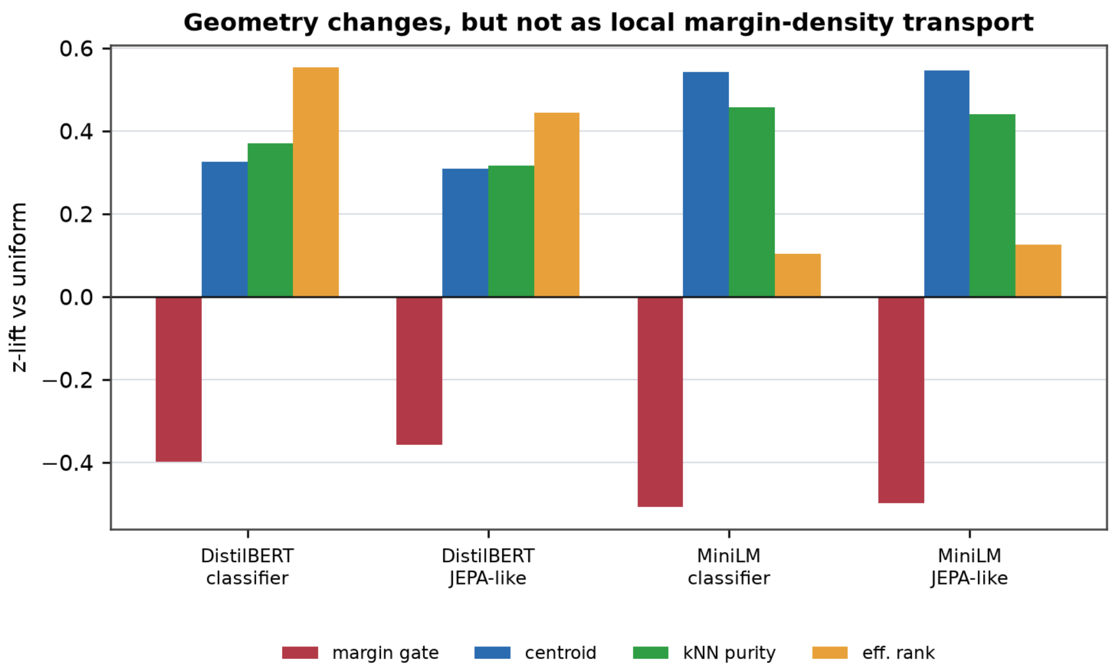

Figure A3. Companion probes reveal non-null geometry. Upweighting increases centroid separation and kNN purity, and often effective rank, even while the registered local semantic-margin gate becomes negative.

The semantic result therefore narrows rather than weakens the main paper. It does not say that semantic loss weighting leaves transformer geometry unchanged. It says the naive class-upweighting analogue in pretrained text encoders is closer to class-level re-centering or cluster sharpening, with slightly lower target F1, than to the local spatial metric-density transport confirmed in the main experiment.

Appendix B. Spatial Methods and Reproducibility

Task Dynamics

The spatial task is 2-D path integration in [0,1]^2 with reflective boundaries. Initial positions are sampled uniformly from [0.1,0.9]^2, headings uniformly from [0,2pi), and headings are updated each step by Gaussian angular noise with standard deviation 0.4. The agent advances at speed

0.06 for T=20 steps. Targets are softmax-normalized Gaussian place codes over a 16x16 grid with sigma_p=0.09.

Loss, Normalization, and Training

The loss is KL divergence to the place-code target, multiplied by w_c(x)=1+6 exp(-||x-c||^2/(2*0.12^2)) for the concern condition. Each batch's weights are divided by their batch mean before the KL multiplication. Uniform controls set w=1 and are matched by architecture, seed, and registered evaluation location. All models use Ng=256 latent units, L2-normalized latent states, readout noise standard deviation 0.15, AdamW with learning rate 1e-3 and weight decay 1e-4, batch size 128, 4,000 training steps, and 1,024 evaluation trajectories.

Architecture Table

Family	Architecture details
RNN	Initial place-code projection to Ng, ReLU RNNCell update from velocity, linear noisy place readout.
Transformer	Velocity and initial-place projections, learned positional embeddings, two causal Transformer encoder layers, four heads, GELU activations, no dropout, and feed-forward width 4Ng.
JEPA-style	MLP place encoder, MLP latent predictor conditioned on velocity, stop-gradient target encoder, and auxiliary target-latent loss with weight 0.5.

Table B1. Spatial model families used in the confirmatory sweep.

Estimator and Bootstrap

The latent metric estimator averages latent states into a 16x16 grid, computes central differences at interior bins, converts log neighbor-stretch to within-model z-scores, then averages a radius-2 neighborhood around the registered target. Bootstrap resampling uses the seed x registered-location reward-control pair as the unit. This is the unit preserved in data/paper_b/reward_location_sweep_2026_07_02_rows.csv.

Compute and Aggregate Data Format

The confirmatory spatial sweep ran on Modal H200/H100 workers and trained 1,920 models: three architectures, 64 seeds, nine concern locations, and one matched uniform control per architecture/seed. The committed CSV snapshot has one row per model-condition-location record with architecture, condition, seed, reward_x, reward_y, final loss, coverage, primary stretch statistics, companion area-density statistics, rank, peak error, top-10 center-of-mass error, and reward-field correlation. The PDF itself is regenerated from committed aggregate numbers; the CSV reproduction script independently recomputes the reported bootstrap summaries.

References

Bennett, W. R. Spectra of quantized signals. Bell System Technical Journal (1948).

Gersho, A., Gray, R. M. Vector Quantization and Signal Compression. Kluwer Academic Publishers (1992).

Cover, T. M., Thomas, J. A. Elements of Information Theory. Wiley (1991).

Amari, S. Natural gradient works efficiently in learning. Neural Computation (1998).

Ganguli, D., Simoncelli, E. P. Efficient sensory encoding and Bayesian inference with heterogeneous neural populations. Neural Computation (2014).

Kriegeskorte, N., Mur, M., Bandettini, P. A. Representational similarity analysis - connecting the branches of systems neuroscience. Frontiers in Systems Neuroscience (2008).

Arvanitidis, G., Hansen, L. K., Hauberg, S. Latent space oddity: on the curvature of deep generative models. ICLR (2018).

Kornblith, S., Norouzi, M., Lee, H., Hinton, G. Similarity of neural network representations revisited. ICML (2019).

Wei, X.-X., Stocker, A. A. A Bayesian observer model constrained by efficient coding can explain 'anti-Bayesian' percepts. Nature Neuroscience (2015).

Cueva, C. J., Wei, X.-X. Emergence of grid-like representations by training recurrent neural networks to perform spatial localization. ICLR (2018).

Banino, A. et al. Vector-based navigation using grid-like representations in artificial agents. Nature 557 (2018).

Sorscher, B., Mel, G. C., Ganguli, S., Ocko, S. A. A unified theory for the origin of grid cells through the lens of pattern formation. NeurIPS (2019).

Gardner, R. J. et al. Toroidal topology of population activity in grid cells. Nature 602 (2022).

Ocko, S. A., Hardcastle, K., Giocomo, L. M., Ganguli, S. Emergent elasticity in the neural code for space. PNAS 115 (2018).

Boccara, C. N. et al. The entorhinal cognitive map is attracted to goals. Science (2019).

Butler, W. N., Hardcastle, K., Giocomo, L. M. Remembered reward locations restructure entorhinal spatial maps. Science (2019).

Vaswani, A. et al. Attention is all you need. NeurIPS (2017).

Assran, M. et al. Self-supervised learning from images with a joint-embedding predictive architecture. CVPR (2023).

LeCun, Y. A path towards autonomous machine intelligence. OpenReview (2022).

Lang, K. NewsWeeder: Learning to filter netnews. ICML (1995).

Devlin, J., Chang, M.-W., Lee, K., Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL (2019).

Sanh, V., Debut, L., Chaumond, J., Wolf, T. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv:1910.01108 (2019).

Wang, W. et al. MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. NeurIPS (2020).